Chair of Mathematical Finance
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Financial Mathematics 1
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Exercise sheet 14

The exercise sheet will be discussed on February 4/5, 2026. For details, see the course's Moodle announcements. You should try to solve the exercises at home before the exercise.

Definition (α-quantile)**.** Recall the general definition of the *non-centered portfolio Value-at-Risk* for the *level* $\alpha \in (0,1)$:

$$\begin{aligned}\mathrm{VaR}_0(\alpha, \boldsymbol{x}) &:= \inf\{m \in \mathbb{R} \ : \ \mathbb{Q}(\Delta V(\boldsymbol{x}) + m < 0) \leq \alpha\} \\ &= \inf\{l \in \mathbb{R} \ : \ \mathbb{Q}(-\Delta V(\boldsymbol{x}) \leq l) \geq 1 - \alpha\}.\end{aligned}$$

Particularly, VaR_0 has two interpretations: First, it is the infimal monetary amount which has to be added to a position such that a loss occurs with a probability less or equal to α. Second, it is the $(1-\alpha)$-quantile of the corresponding loss distribution.

Theorem (Exchanging the Order of Differentiation and Integration, see [1, Thm. 6.28])**.** *Let $(\Omega, \mathcal{F}, \nu)$ be a measurable space and $I \subseteq \mathbb{R}$ be an open, non-trivial interval and $f : \Omega \times I \to \mathbb{R}$ a mapping with the properties:*

i) for every $x \in I$, we have $(\omega \mapsto f(\omega, x)) \in L^1(\nu)$,

ii) for almost every $\omega \in \Omega$, the mapping $I \to \mathbb{R}$, $x \mapsto f(\omega, x)$ is differentiable and we denote this (partial) derivative by f',

iii) there exists a non-negative $h \in L^1(\nu)$ with $|f'(\cdot, x)| \leq h$ ν-a.e. for all $x \in I$.

Then, for all $x \in I$, we have $f'(\cdot, x) \in L_1(\nu)$ and the function $F : I \to \mathbb{R}$, $x \mapsto \int f(\omega, x)\nu(\mathrm{d}\omega)$ is differentiable with derivative

$$F'(x) = \int f'(\omega, x)\nu(\mathrm{d}\omega).$$

Exercise 14.1

Assume a market consisting of N risky assets with return vector $\boldsymbol{R}$. Assume that $\boldsymbol{R}$ has a continuous distribution (w.r.t. the N-dimensional Lebesgue measure) and that $\mathbb{E}[|R|] < \infty$. Furthermore, assume a riskless asset with return $r > 0$ and an investor with initial wealth $v > 0$ and linear investment constraints

$$X = \left\{\boldsymbol{x} \in \mathbb{R}^N \ : \ A\boldsymbol{x} \leq \boldsymbol{b} \text{ and } A_{\mathrm{eq}}\boldsymbol{x} = \boldsymbol{b}_{\mathrm{eq}}\right\} \neq \emptyset.$$

Note that the *loss* of a portfolio $\boldsymbol{x}$ with initial investment v is

$$-\Delta V(\boldsymbol{x}) = -(v(1 + \boldsymbol{R}(\boldsymbol{x})) - v) = -v(1 + \boldsymbol{R}(\boldsymbol{x}) - 1) = -v\big[r + \boldsymbol{x}^\top(\boldsymbol{R} - r\mathbf{1})\big]$$

and consider, for fixed $\alpha \in (0,1)$, the following two optimization problems:

$$\min_{\boldsymbol{x} \in X} \mathrm{CVaR}(\alpha, \boldsymbol{x}) \tag{$P_{\mathrm{CVaR}(\alpha)}$}$$

$$\min_{\boldsymbol{x} \in X, \zeta \in \mathbb{R}} \mathrm{lCVaR}(\alpha, \boldsymbol{x}, \zeta), \tag{$P_{\mathrm{lCVaR}(\alpha)}$}$$

with

$$\mathrm{lCVaR}(\alpha, \boldsymbol{x}, \zeta) = \zeta + \frac{1}{\alpha}\mathbb{E}\Big[(-\Delta V(\boldsymbol{x}) - \zeta)^+\Big].$$

We show in this exercise, which follows the arguments of [2], that both problems are equivalent and that the second problem is convex. In particular, if $\boldsymbol{x}^*$ is a solution of $(P_{\mathrm{CVaR}(\alpha)})$, then $(\boldsymbol{x}^*, \mathrm{VaR}_0(\alpha, \boldsymbol{x}^*))$ is a solution of $(P_{\mathrm{lCVaR}(\alpha)})$ and if $(\boldsymbol{x}^*, \zeta^*)$ is a solution of $(P_{\mathrm{lCVaR}(\alpha)})$, then $\boldsymbol{x}^*$ is a solution of $(P_{\mathrm{CVaR}(\alpha)})$. Hence, for lack of a better term, we call the second problem, $(P_{\mathrm{lCVaR}(\alpha)})$, the *lifted CVaR problem*.

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a) Show that $\mathrm{lCVaR}(\alpha, \boldsymbol{x}, \zeta)$ is a convex function of $(\boldsymbol{x}, \zeta)$.

b) Fix $\boldsymbol{x} \in X$ and consider the following (auxiliary) optimization problem:

$$\min_{\zeta \in \mathbb{R}} \mathrm{lCVar}(\alpha, \boldsymbol{x}, \zeta). \tag{$P_{\mathrm{AUX}(\alpha,\boldsymbol{x})}$}$$

Show that the objective function of this optimization problem is convex and continously differentiable, that the set of solutions form a non-empty closed interval with left end-point $\mathrm{VaR}_0(\alpha, \boldsymbol{x})$, and that the objective function evaluated at any solution equals $\mathrm{CVaR}(\alpha, \boldsymbol{x})$. *Hints: (1) You may use without proof that exchanging the order of integration and differentiation is applicable to* $\mathrm{lCVaR}(\alpha, \boldsymbol{x}, \zeta)$ *as a function of* ζ *and that the objective function of* $(P_{\mathrm{AUX}(\alpha,\boldsymbol{x})})$ *is continuously differentiable (if you want, you can also prove these facts). (2) You may use without proof that* $\boldsymbol{R}$ *having a continuous distribution implies that* $-\Delta V(\boldsymbol{x})$ *has a continuous distribution (w.r.t. the one-dimensional Lebesgue measure) and consequently that for each* $\alpha \in (0, 1)$ *there exists (at least one)* $\zeta \in \mathbb{R}$ *such that*

$$\mathbb{Q}(-\Delta V(\boldsymbol{x}) \leq \zeta) = 1 - \alpha.$$

c) Show that

$$\min_{\boldsymbol{x} \in X} \mathrm{CVaR}(\alpha, \boldsymbol{x}) = \min_{\boldsymbol{x} \in X, \zeta \in \mathbb{R}} \mathrm{lCVaR}(\alpha, \boldsymbol{x}, \zeta)$$

and in particular, that for every solution $\boldsymbol{x}^*$ of $(P_{\mathrm{CVaR}(\alpha)})$, $(\boldsymbol{x}^*, \mathrm{VaR}_0(\alpha, \boldsymbol{x}^*))$ is a solution of $(P_{\mathrm{lCVaR}(\alpha)})$ and for every solution $(\boldsymbol{x}^*, \zeta^*)$ of $(P_{\mathrm{lCVaR}(\alpha)})$, $\boldsymbol{x}^*$ is a solution of $(P_{\mathrm{CVaR}(\alpha)})$.

References

[1] Klenke, A. Wahrscheinlichkeitstheorie. *Springer Spektrum, Berlin, Heidelberg*, Ed. 3, 2013.

[2] Rockafellar, R. T., & Uryasev, S. Optimization of conditional value-at-risk. *Journal of risk*, 2, 21–42, 2000.